

Pumping station

PS.W

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions.

These installation and operating instructions describe Grundfos PUST PS.W.

Sections 1-5 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 6-10 give important information about the product, as well as information on service, fault finding and disposal of the product.

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury.



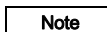
Warning

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.



Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note

Notes or instructions that make the job easier and ensure safe operation.

2. Safety

Receiving the product



Warning

Do not stand under or close to a suspended load.

Warning

Before lifting the pit, make sure that the lifting equipment is capable of lifting this load.



Make sure that the lifting bracket is tightened before attempting to lift the pit.

Tighten if necessary. Carelessness during lifting or transportation of the pit may cause injury to persons or damage to the pit.

Installing the product

Warning

Installation of pits must be carried out by an authorised person in accordance with local regulations.



Work in or near wastewater pits must be carried out according to local regulations.

Warning

The lifting equipment used for lifting the pump must be rated for the weight of the pump, approved and maintained according to local regulations.



Warning

Chains supplied by Grundfos are marked with maximum load and production date.
The maximum load must not be exceeded.
Maintenance according to local regulations. We recommend that you check chains and shackles supplied by Grundfos at least once per year for cracks, corrosion or other irregularities. If any defects are found, replace the chain or shackles.



Covers are not approved for traffic unless stated otherwise.



Be aware that covers are adequate closed and locked to prevent unwanted access.
An open pit must never be left unattended.

Warning

The electrical installation must be carried out by an authorised person in accordance with local regulations.

**Warning**

The pump or pump controller must be connected to an external emergency stop.
If a power supply circuit breaker is used as emergency stop, it must fulfil EN 60204-1, 10.8.4.

**Warning**

Before starting work on the pump or valves, make sure that the fuses have been removed or the main switch has been switched off.
Make sure that the power supply cannot be accidentally switched on.

**Warning**

The control cabinet and internal pipe work must be earthed.

**Servicing the product****Warning**

Do not enter and work inside PS.W04-06-08-10.

**Warning**

Before entering PS.W 1200 - 3000 make sure that the cover and safety grid are locked in open position and that the pit is ventilated according to local regulations. If not, do not enter the pit.

**Warning**

When entering the pit, wear safety harness and use proper ladder and equipment for lifting persons up from the pit.
All work in pits must be carried out according to local regulations and supervised by at least one person outside the pumping station.

**Warning**

When work is going on in an open, or in the proximity of an open pumping station or valve chamber, place warning signs and safety barriers around the pit in order to avoid that persons fall into the pit. The warning signs must be visible from all directions.

**Warning**

If the top opening is up to Ø1000, normal precautions will be sufficient. Openings larger than Ø1000 must be equipped with safety barriers or other safety measures.

**Warning**

Pumps can be lifted by means of a crane, using lifting points.
Slings or chains, approved for lifting must be used.

**Warning**

Gloves and other suitable personal protection equipment must be used in accordance with local regulations.
Observe local regulations on exposure to wastewater.

**Warning**

Make sure that the escaping water does not cause injury to persons or damage to the equipment.

**Warning**

Before starting work on the non-return valves, make sure that the fuses have been removed or the main switch has been switched off.
Make sure that the power supply cannot be accidentally switched on.

**Warning**

Make sure that the isolating valves cannot be accidentally opened.

**Warning**

If a pump has been used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.



3. Receiving the product

3.1 Transporting and handling the product

Note

When transporting and handling the pit at low temperatures, take into consideration that the pit impact resistance is reduced.

Precautions during transport and handling:

- Do not dump the pit from the lorry.
- Use approved straps of textile or similar materials when lifting the pit on to or down from a lorry or moving it on the construction site.
- Handle and lift the pit according to local regulations.
- When lifting the pumping station with a strap, use the ribs on the pumping station.
- Do not drag the pit over the ground.
- Point loads must not occur.
- The pit must not be exposed to sharp edges.
- Fix the pit securely during transport.
- When the pit is placed on the ground, make sure that the ground is even.

Note

The weight of a PS.W pit can vary from 50 to maximum 2500 kg.

Caution

If the pit is equipped with a lifting point, use it during handling.

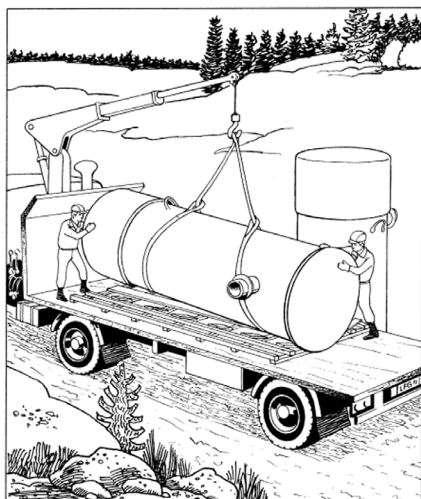


Fig. 1 Lifting the pit from a lorry

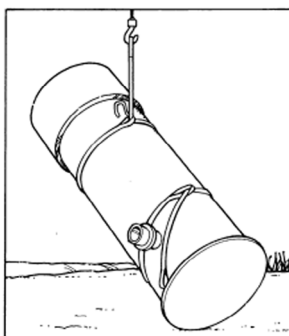


Fig. 2 Placing the pit in upright position by straps

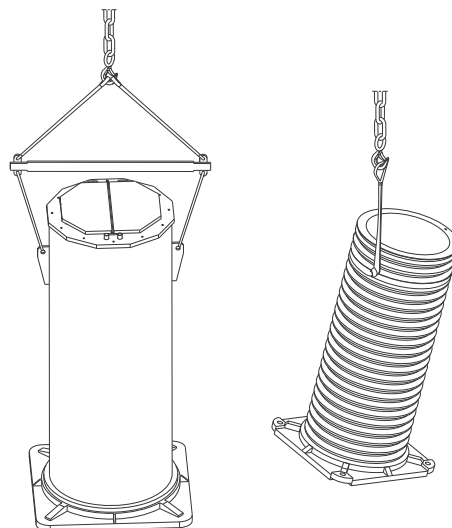


Fig. 3 Lifting the pit

Caution

Support for the crane must be placed in a suitable distance from the pit to avoid collapse of pit hole.

Observe local regulations.


Warning

Do not stand under or close to a suspended load.


Warning

Before lifting the pit, make sure that the lifting equipment is capable of lifting this load.

Make sure that the lifting bracket is tightened before attempting to lift the pit.

Tighten if necessary. Carelessness during lifting or transportation of the pit may cause injury to persons or damage to the pit.

3.2 Storage

Caution
Warning

If the ambient temperature exceeds 30 °C, store the pit in upright position to avoid deformation.

3.3 Inspecting the product

After transportation and before installation, the pumping station must be inspected by the customer.

The inspection must include the following:

- Check the pumping station for transport damage, etc. Contact the transporter immediately if you detect any damage.
- Check that the products delivered correspond to the order.
- Check the positions and sizes of fittings.
- Check and retighten all bolted connections, as they may have become loose during transport.
- Check that the valves are open.
- Check other equipment such as venting pipes.

TM06 1232 1914

TM06 1233 1914

TM06 3630 0715

4. Installing the product

4.1 Preparing the foundation



Warning

Installation of pits must be carried out by an authorised person in accordance with local regulations.

Work in or near wastewater pits must be carried out according to local regulations.

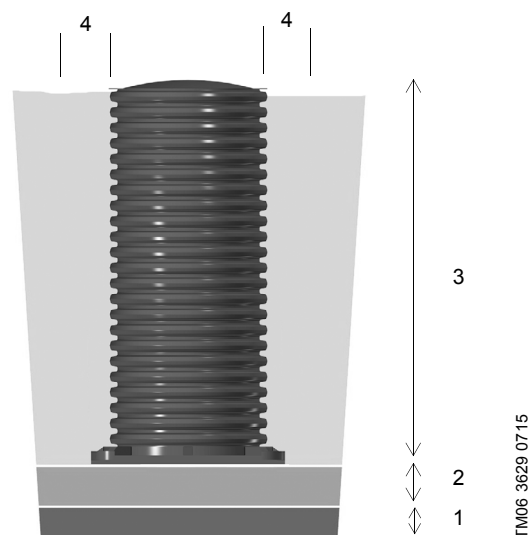


Fig. 4 Schematic installation drawing

Pos.	Description
1	Foundation layer
2	Bedding layer
3	Backfill compacted in layers of maximum 30 cm
4	Distance of 50 cm from cover where heavy loads must not occur

Caution

The location of the pit must be selected so that its installation does not damage other equipment. And the other equipment must not damage the buried pit.

4.1.1 Foundation layer

- The pit must be placed on a foundation layer if soil analyses and information about pit load show that the soil is not capable of bearing the weight. The bedding layer on which the pit is placed is not regarded as a foundation layer.
- The foundation layer can be made after the excavation by laying a stable layer of suitable gravel or similar material and compacting it in layers of maximum 30 cm (corresponding to approx. 20 cm after compaction). Such a foundation layer is also required if the excavation has become too deep by mistake.

4.1.2 Bedding layer

- The material for the bedding layer must be compactable and composed so that neither its properties nor the compaction will cause damage to the pit.
- Grains larger than 16 mm must not be present.
- The content of grains between 8 and 16 mm must not exceed 10 %.
- The material must not be frozen.
- Sharp flint etc. must not be present.
- The bedding layer must have a thickness of 10 cm.

4.2 Installing the pit

Before lowering the pit into position, retighten various connections, as they may have become loose during the transport.

Note

Check the pit for external damage before lowering it into position. When the pit has been installed, Grundfos cannot be held responsible for possible damage.

Caution

Mounting on or drilling in the bottom plate is not allowed.

Any damages may compromise the strength of the bottom plate which can affect the complete pit.

Caution

Do not use heavy equipment close to the pit, to avoid collapse of the pit hole or deformation of the pit.

- Make sure to position the pit correctly so the flanges are in the right direction according to inlet and outlet pipes.
- Position the pit on the bedding layer.
- Check that the pit is vertical.

Groundwater must not rise above the bedding layer until backfilling has been completed.

4.3 Backfill

- The backfill must provide sufficient support of the pit on all sides and ensure that the load can be transferred without detrimental point impacts or similar impacts.
- The backfill must meet the same requirements as those for the bedding layer.
- The backfilling must be carried out so that the pit is not damaged or deformed.
- The backfill must be compacted in layers of maximum 30 cm (corresponding to approx. 20 cm after compaction).

Compact the backfill under the inlet and outlet pipes properly so that they are not exposed to downward loads when the backfill settles. See fig. 5.

Note

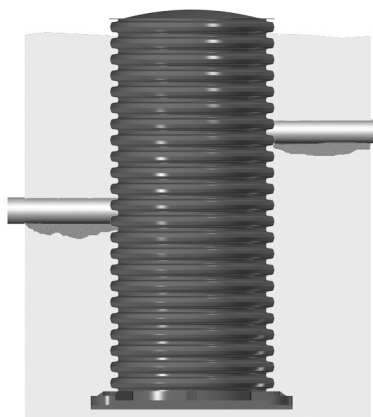


Fig. 5 Insufficient compaction under pipes

During backfilling, the cover must be fitted on the pit to ensure that the pit is not deformed (becomes oval).

Note

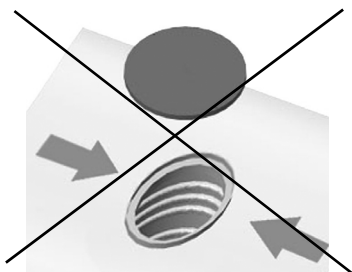
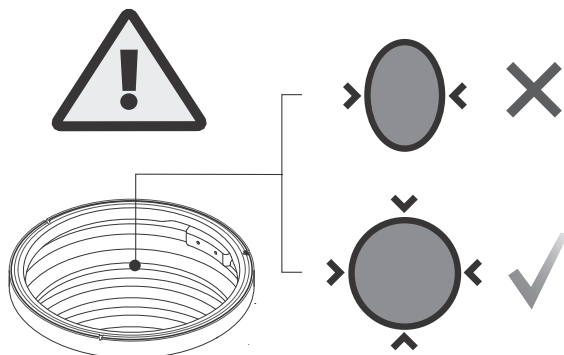
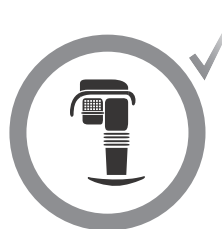


Fig. 6 Oval pit after compaction



Caution

If the hole for the pit is drilled, pay special attention to the compaction of the backfill to prevent uplift.



Max.
25 kN

Max.
150 kg

TM06 3602 0615

Caution

Use mechanical compacting equipment and compact the backfill to 98-100 % proctor density.

4.4 Uplift prevention

The pit design ensures that the pit is protected against uplift if installed correctly. Requirements for uplift prevention depend on the geotechnical conditions. They should be determined by the engineer or contractor and are outside Grundfos' area of responsibility.

Caution

The above requirements are minimum requirements. Observe local regulations.

4.5 Installing the valve chamber

Follow the same instructions as for installation of the pit. See section 4. [Installing the product.](#)

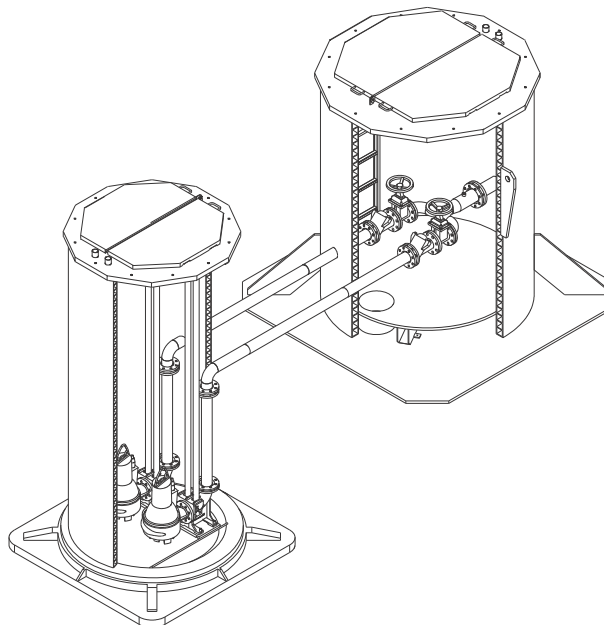


Fig. 7 Valve chamber

TM06 3628 0715

TM02 9361 2404

TM06 3603 0615

TM06 3496 0615

4.6 Connecting the pipes

Holes for inlet pipe, electrical connection and venting are to be drilled on site.

Caution

Placement of inlet and outlet pipe connections must be carried out in accordance with local regulations.

1. Mark the pit where the inlet is to be.
2. Drill the marked spot with a hole saw.
3. Remove all burrs from the hole.

4.6.1 Drilling the inlets

Figures 8 and 9 show which hole saw to use for the various pumping station diameters depending on the difference in the wall thickness.

PS.W.04 - PS.W.10

Diameter [mm]	Wall thickness [mm]	Profile width [mm]
Ø	A	B
400	30	48
600	44	66
800	57	88
1000	72	106

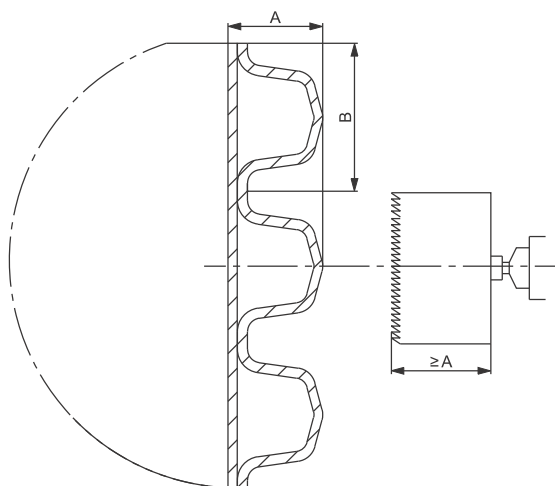


Fig. 8 Wall profile PS.W 400-1000

PS.W.12 - PS.W.30

Wall thickness varies for SN2-SN4-SN8

Diameter [mm]	X [mm]
1250-3000	50-170

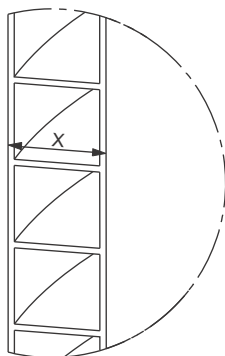


Fig. 9 Wall profile PS.W 1250-3000

4.7 Location of inlet

The pit inlet must not be located within the area shown in fig. 10, as it will disturb the function of the float switches.

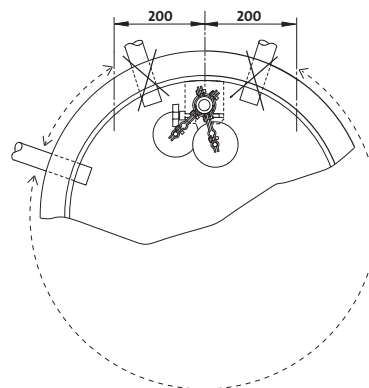


Fig. 10 Location of inlet relative to level control

Caution

The pit inlet must not be located within the area shown in figs 11 and 12 as it will interfere with piping and valves.

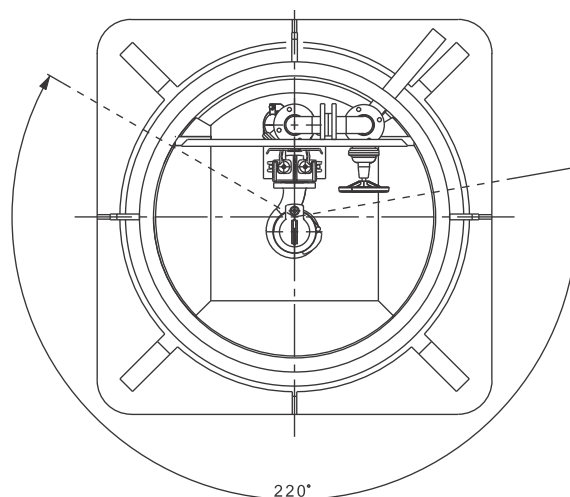


Fig. 11 Location of inlet relative to pipes with one pump

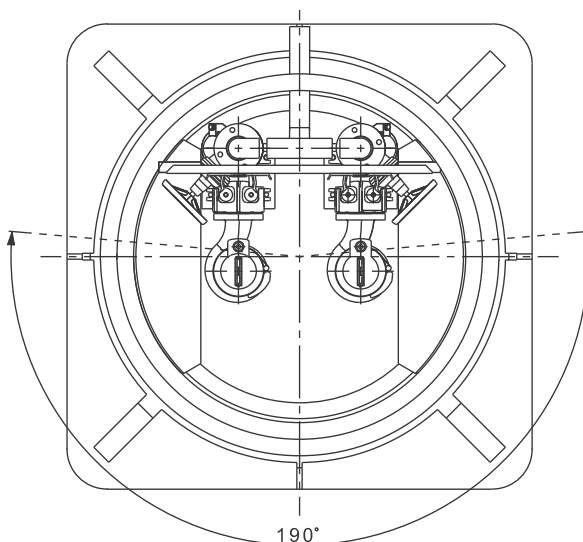


Fig. 12 Location of inlet relative to pipes with two pumps

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TM06 3613 0615

TM06 3612 0615

TM06 3605 0715

TM06 3610 0615

Caution

The pit inlet must not be located within the area shown in fig. 13 as it will destroy the function of the inclined bottom plates.

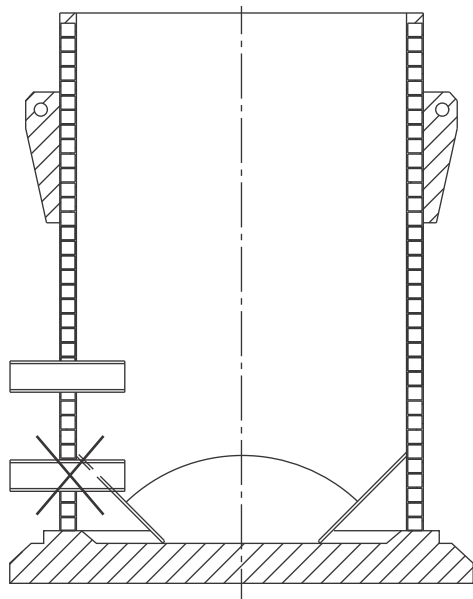


Fig. 13 Location of inlet relative to inclined bottom plates

4.8 Fitting the sleeve

1. Fold the sleeve and insert it into the hole from the outside of the pit.
2. Unfold the sleeve so that it lies evenly in the hole.
3. **LM50/100 sleeves:** Pull the sleeve outwards until its collar touches the inside of the pit all around the hole.
L965 sleeves: Pull the sleeve outwards until its collar touches the inside of the pit at the positions 3 and 9 o'clock. See fig. 16.
4. Lubricate the inside of the sleeve with a water-based lubricant.
5. Chamfer the pipe and press it home.

LM50/100 sleeves: Push the inlet pipe through the sleeve. See fig. 14.

Note

As it must be possible to pull the pump up through the pit, the inlet pipe must not project beyond the sleeve by more than 5 cm.

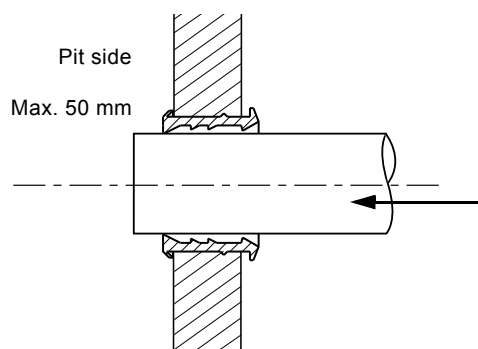


Fig. 14 Pipe with LM50/100 sleeve, seen from side

L965 sleeves: Push the inlet pipe into the sleeve and against the stop. See figs 15 and 16.

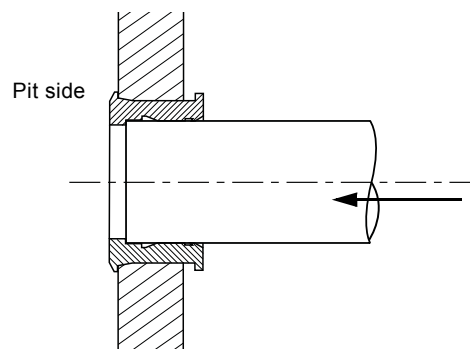


Fig. 15 Pipe with L965 sleeve, seen from side

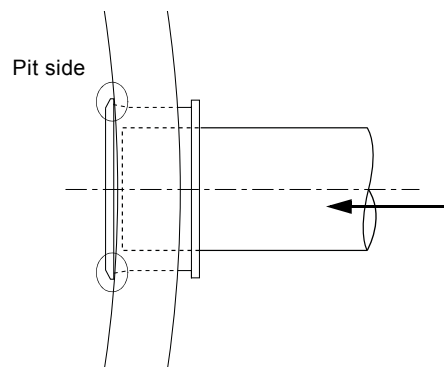


Fig. 16 Pipe with L965 sleeve, seen from top

4.9 Installing the pump(s)

Some versions come without the pump(s) installed. For installation and startup of the pump, see the installation and operating instructions for the pump.

Caution

Lower the pump carefully into the pit in order to avoid damage to pump and pit.

4.10 Pipework with flange connections

Note

Loosen all bolts of the flange and retighten them when the pit has been installed. This will prevent stress in the pipes.

4.11 Fitting the chain

For pumps on auto coupling, we recommend that you fit the chain in the foremost lifting eye of the lifting bracket. The number of lifting eyes depends on the pump.



Fig. 17 Chain fitted to pump

TM02 9341 2404



Warning

The lifting equipment used for lifting the pump must be rated for the weight of the pump, approved and maintained according to local regulations.



Warning

Chains supplied by Grundfos are marked with maximum load and production date. The maximum load must not be exceeded. Maintenance according to local regulations. We recommend that you check chains and shackles supplied by Grundfos at least once per year for cracks, corrosion or other irregularities. If any defects are found, replace the chain or shackles.

4.12 Installing covers



Covers are not approved for traffic unless stated otherwise.

Be aware that covers are adequate closed and locked to prevent unwanted access.

An open pit must never be left unattended.

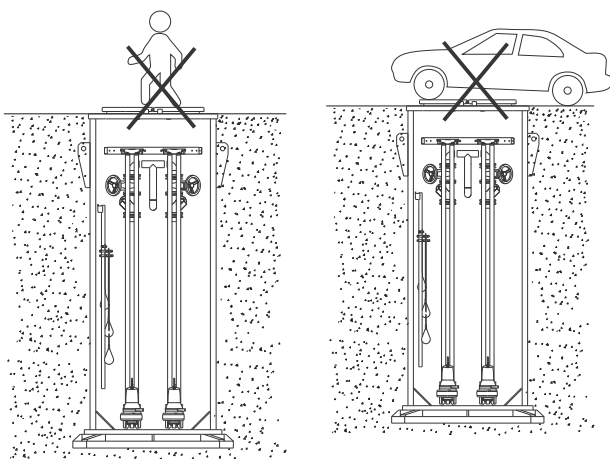


Fig. 18 Not approved for pedestrians and vehicles

TM06 0113 4913 - TM06 0114 4913

4.13 Installing a cover approved for traffic

Caution *The maximum angle of inclination is 25 °.*

The foundation and bedding layer for the cover must be of such nature so the surface can carry the load that the concrete cover is rated. Observe local regulations.

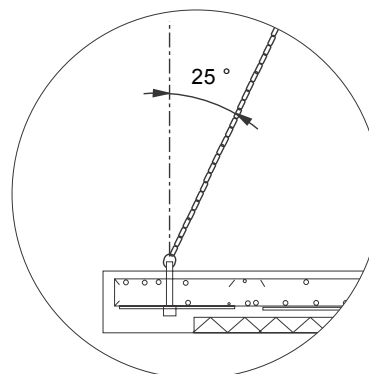


Fig. 19 Maximum angle of inclination

TM06 3498 0615

Lower the cover onto the pit when the backfill has been compacted to the top level of the pit. See fig. 21. After installation, the air space between the cover and pit must be 150 mm. See fig. 21.

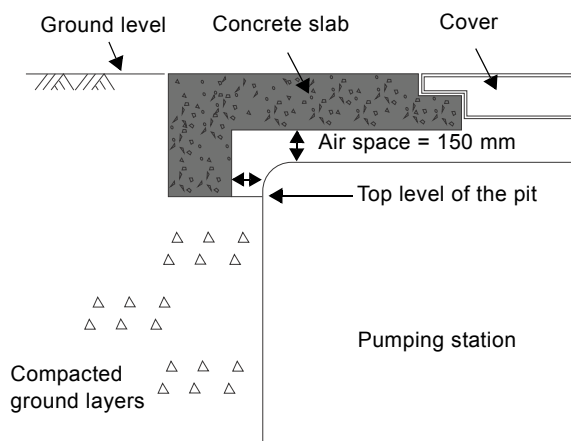


Fig. 20 Cover approved for traffic

TM06 3575 0615

4.14 Venting pipe

If the cover is not pre-installed with a venting pipe, we recommend that you provide the pit with a venting pipe (accessory).

For installation, see section [4.6 Connecting the pipes](#).

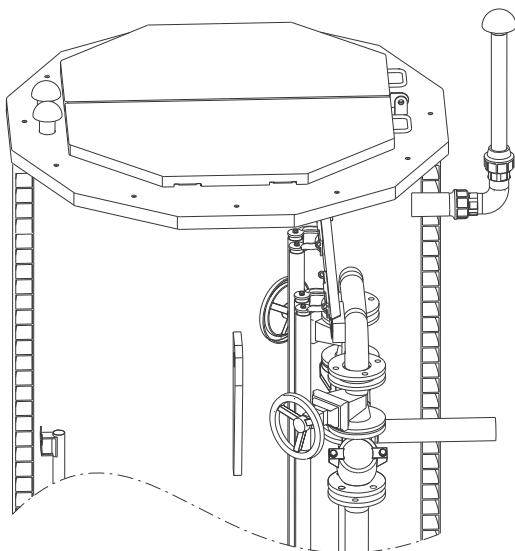


Fig. 21 Pit with venting pipe

4.15 Cables

The cables to the level switches and the pump can be led into the pit through a cable entry in the side of the upper region of the pit. See fig. [22](#).

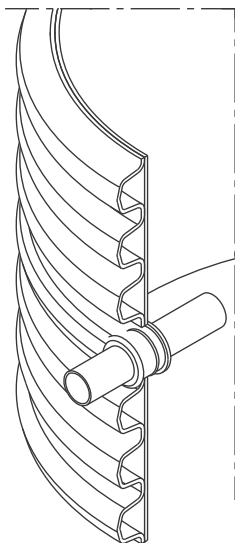


Fig. 22 Pipe for cable entry

When dismantling or assembling the pump, make sure not to pinch or damage the cables.

Caution

When the pump and cables have been mounted, the cables must be suspended in such a way that any stress is relieved from the cables.

4.16 Electrical installation

See the installation and operating instructions for the pump and the pump controller.



Warning

The electrical installation must be carried out by an authorised person in accordance with local regulations.



Warning

The pump or pump controller must be connected to an external emergency stop.

If a power supply circuit breaker is used as emergency stop, it must fulfil EN 60204-1, 10.8.4.



Warning

Before starting work on the pump or valves, make sure that the fuses have been removed or the main switch has been switched off.

Make sure that the power supply cannot be accidentally switched on.

Caution

All cable inlets must be sealed properly.

Note

Do not install Grundfos control boxes and the free end of the supply cable inside the pumping station.

Caution

Pump cables must be connected as described in the installation and operating instructions of the pump.



Warning

The control cabinet and internal pipework must be earthed.

Caution

Cables must be handled with care to avoid damages on the cable insulation. Excessive cables must be suspended in cable hooks.

Caution

Cables must be protected at all the times to prevent water from entering the control box capillary.

4.17 Installing the pump controller

See the installation and operating instructions for the pump controller.

4.18 Installing the level controllers

4.18.1 Float switches

If float switches have been selected, they can be fitted on a tube which can be lifted out of the pit. This ensures easy adjustment of float switches.

Note

Note that the bottom float switch (stop) must stop the pump before the level in the pit falls below the minimum level of the pump. See the installation and operating instructions for the pump.

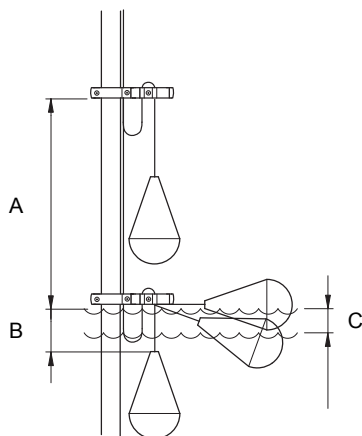


Fig. 23 Adjusting the float switches

A	Min. 300 mm
B	50 to 100 mm
C	Deactivation range 110 mm

Caution

Distance B must not be too big as otherwise the float switch may get stuck in other parts of the installation.

4.18.2 Electrodes

If electrodes are used for level control, note that the electrodes come in a standard length (1 m). When they are installed, their lengths must therefore be adjusted so that they are activated at the correct levels.

If possible, bend the electrodes away from each other so that paper and similar items cannot get stuck between the electrodes and disturb the signal.

When they have been shortened to the desired length, strip 10 to 15 mm of the lower end of the electrodes.

The electrodes come with a 10 m cable.

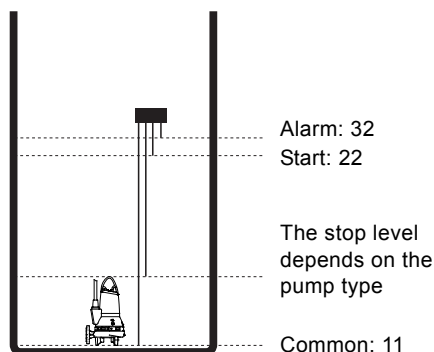


Fig. 24 Electrodes in a pit with one pump

TM02 8962 1204

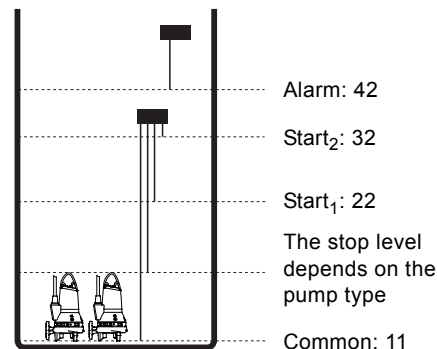


Fig. 25 Electrodes in a pit with two pumps

TM02 8963 1204

4.19 Air bells

Air bells are pulse generators for the pressure switches of the pump controller and function by means of pneumatic pressure changes. The pressure changes when the liquid level changes in the air bell. The air bells are connected to the pressure switches of the controller by means of pressure hoses.

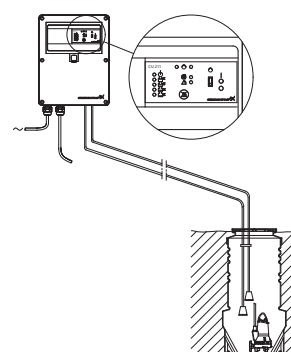


Fig. 26 Air bells in a pit with one pump

TM01 9345 2404

4.20 Pressure transducer

If a pressure transducer is used, it must be installed in a protective pipe in order to prevent contamination and deposits.

5. Pump control

Note

If the pit is equipped with an AutoADAPT controlled pump, an external level controller is not needed.

Caution

Check that the pump control is designed for the pumps to be installed.

When installing the level switches, observe the following points:

- To prevent air intake and vibrations in the pump, the stop level switch must be fitted in such a way that the pump is stopped before air is sucked into the pump.
- In the case of 1-pump operation, the start level switch should be installed in such a way that the pump is started at the required level; however, the pump must always be started before the liquid level reaches the lower edge of the bottom inlet pipe.
- In the case of 2-pump operation, the start level switches must be installed in such a way that pump 2 is started before the liquid level reaches the lower edge of the bottom inlet pipe, and so that pump 1 is started correspondingly earlier.
- The high-level alarm switch must always be installed about 10 cm above the start level switch; however, the alarm must always be given before the liquid level reaches the inlet pipe.

For further settings, see the installation and operating instructions for the pump controller selected.

5.1 Start and stop levels

The effective volume of the pit must be so large that the number of starts does not exceed the maximum permissible number. See the installation and operating instructions for the pump.

6. Product introduction

Grundfos pumping stations are prefabricated pumping stations for collection and pumping of wastewater.

The pump pit is made of PE-HD (polyethylene) or PP (polypropylene) and comes with pipes and valves fitted.

The pipework is made of PE, stainless steel or galvanized steel depending on the location of the pit and the pumped liquid, and thus the pump selected.

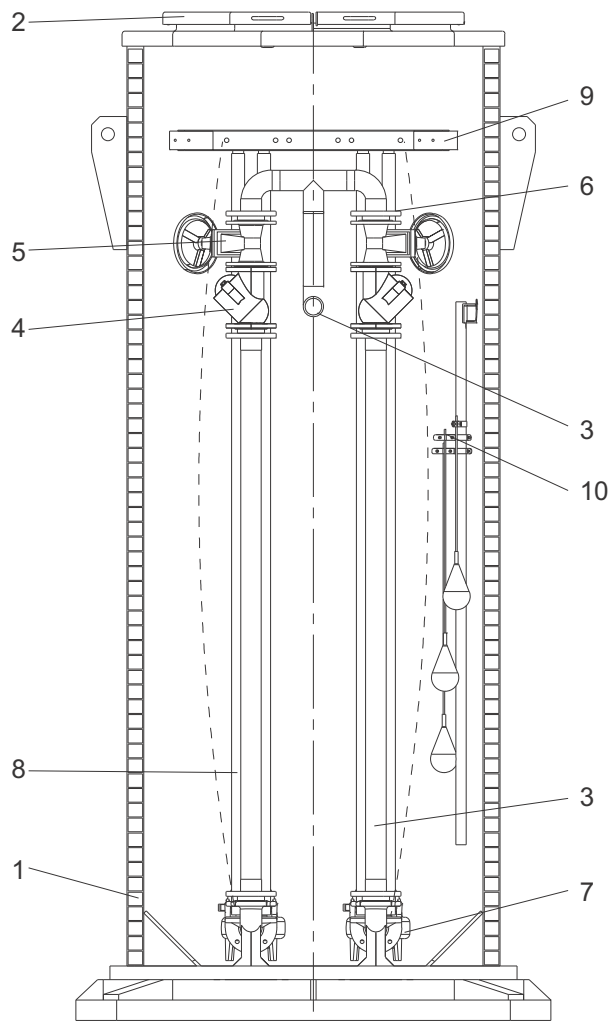


Fig. 27 Example of pumping station

Pos.	Description
1	Pit
2	Cover
3	Pipes
4	Non-return valve
5	Isolating valve
6	Pipe coupling
7	Auto coupling
8	Guide rails
9	Guide rails holders
10	Level control system

7. Applications

Grundfos pumping stations are used for collection and pumping of drainage water, rainwater and wastewater. Wastewater is led into the pumping station. When the liquid in the pumping station reaches the starting level, the pump starts and pumps the liquid further in the system to a sewage treatment plant or sewer.

The pits are prepared for various pumps depending on the pumped liquid.

7.1 Pumped liquids

- Drainage water
- Rainwater (surface water)
- Wastewater

7.1.1 Liquid temperature

Maximum 40 °C. For higher temperatures, contact Grundfos.

Caution Select the pump on the basis of knowledge of the liquid temperature. See the installation and operating instructions for the individual pumps.

7.1.2 Acids and alkalis

The pumping station is resistant to strong acids and alkalis as well as solvents.

The pumps are supplied with the pump pit and normally capable of withstanding pH values between 4 and 10. In case of doubt, please contact Grundfos.

7.1.3 Density

Maximum 1100 kg/m³.

7.2 CE approval of PS.W

PS.W is CE-approved in accordance with these standards:

- EN 12050-1
- EN 12050-2.

7.3 Identification

7.3.1 Nameplate, PS.W

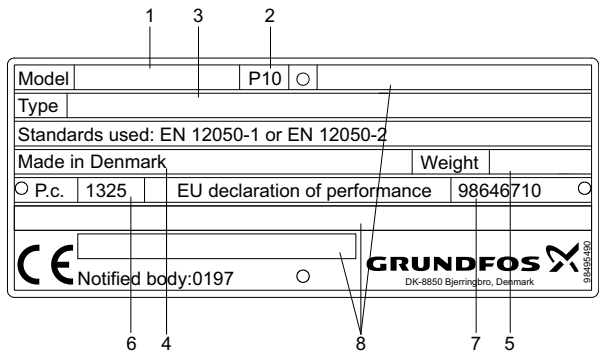


Fig. 28 Nameplate, PS.W

Pos.	Description
1	Product number
2	Production site
3	Type designation
4	Country of origin
5	Weight
6	Production code - YYWW of production
7	Installation and operating instructions, publication number
8	Not filled in

7.3.2 Type key, PS.W

Example	PS.	W.	20.	25.	D.	GC.	SS100	A100.	SEG
Pumping station									
Pit type and material W: Welded PE or PP									
Diameter [mm] xx: x 100 20: 2000 mm									
Depth [mm] xx: x 100 mm 25: 2500									
Number of pumps N: No pump S: 1 pump D: 2 pumps T: 3 pumps F: 4 pumps									
Pipe design DC: Direct outlet, common DS: Direct outlet, separated GC: Goose neck, common VC: Valve chamber CL: Launcher for cleaning pig NV: No valves, separated									
Pipe material and pipe diameter Stainless steel (304 and 316): SS32: DN 32 (1") SS40: DN 40 (1 1/2") SS50: DN 50 (2") SS65: DN 65 (2 1/2") SS80: DN 80 (3") SS100: DN 100 (4") SS125: DN 125 (5") SS150: DN 150 (6") SS200: DN 200 (8")	Galvanized steel: GA40: DN 40 (1 1/2") GA50: DN 50 (2") GA65: DN 65 (2 1/2") GA80: DN 80 (3")	Polyethylene: PE63: D63 mm (2") PE75: D75 mm (2 1/2") PE90: D90 mm (3") PE110: D110 mm (4")							
Installation type Auto coupling: A40: DN 40 pump connection A50: DN 50 pump connection A65: DN 65 pump connection A80: DN 80 pump connection A100: DN 100 pump connection A150: DN 150 pump connection A200: DN 200 pump connection Free-standing pump: S: Free-standing pump Wall-hanging W: Wall-hanging Hook-up H: Hook-up									
Pump type CC: Unilift CC KP: Unilift KP AP12: Unilift AP12.50 AP35: Unilift AP35, Unilift AP12.40 AP50: Unilift AP50 APB: Unilift AP35B, Unilift AP50B SEG: SEG DP/EF: DP (0.6 - 1.5 kW) / EF DP/SL: DP (2.6 kW) / SL1.50.65 / SLV.65.65 SE/SL: SE/SL									

8. Servicing the product

See the installation and operating instructions for the pump and the pump controller.

Note

We recommend that you make all maintenance and service work when the pump is placed outside the pit.



Warning

Do not enter and work inside PS.W04-06-08-10.



Warning

Before entering PS.W 1200-3000, make sure that the cover and safety grid are locked in open position and that the pit is ventilated according to local regulations. If not, do not enter the pit.



Warning

When entering the pit, wear safety harness and use proper ladder and equipment for lifting persons up from the pit.

All work in pits must be carried out according to local regulations and supervised by at least one person outside the pumping station.



Warning

When work is going on in an open, or in the proximity of an open pumping station or valve chamber, place warning signs and safety barriers around the pit in order to avoid that persons fall into the pit. The warning signs must be visible from all directions.



Warning

If the top opening is up to Ø1000, normal precautions will be sufficient. Openings larger than Ø1000 must be equipped with safety barriers or other safety measures.



Warning

Pumps can be lifted by means of a crane, using lifting points.

Slings or chains, approved for lifting must be used.



Warning

Gloves and other suitable personal protection equipment must be used in accordance with local regulations.

Observe local regulations on exposure to wastewater.

Note

If natural light is insufficient, the maintenance staff must use lamps.

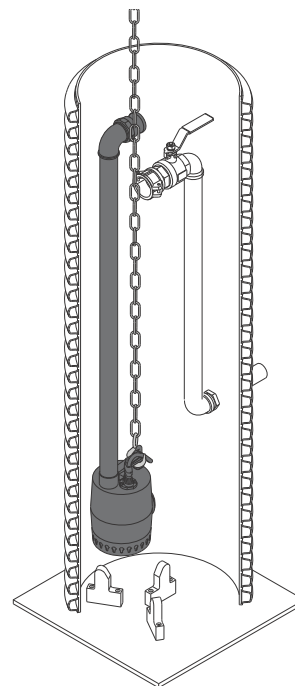


Fig. 29 Lifting a free-standing pump for service

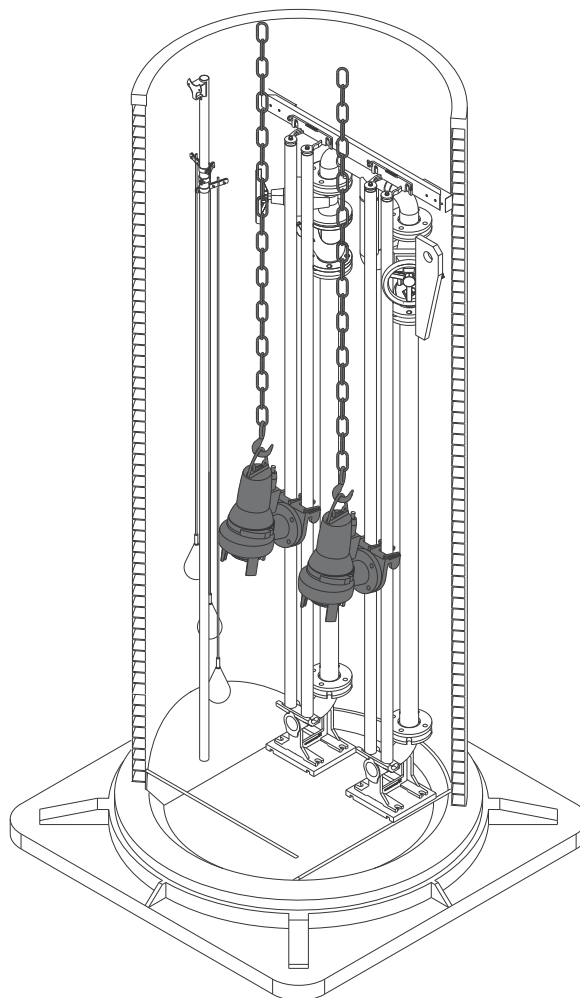


Fig. 30 Lifting pumps on auto coupling for service

TM06 3633 0715

TM06 3634 0715

8.1 Repairing a non-return valve



Warning

Make sure that the escaping water does not cause injury to persons or damage to the equipment.



Warning

Before starting work on the non-return valves, make sure that the fuses have been removed or the main switch has been switched off.

Make sure that the power supply cannot be accidentally switched on.



Warning

Make sure that the isolating valves cannot be accidentally opened.

Caution

When entering the valve chamber, do not step on pipes or valves

Non-return valve placed inside the pumping station

1. Close the isolating valves. See fig. 33
2. Remove the two screws on the non-return valve cover. See fig. 31.
3. Replace worn-out valve balls and clean the valves inside.
4. Close the valve cover and tighten the screws.
5. Open the isolating valves. See fig. 31.

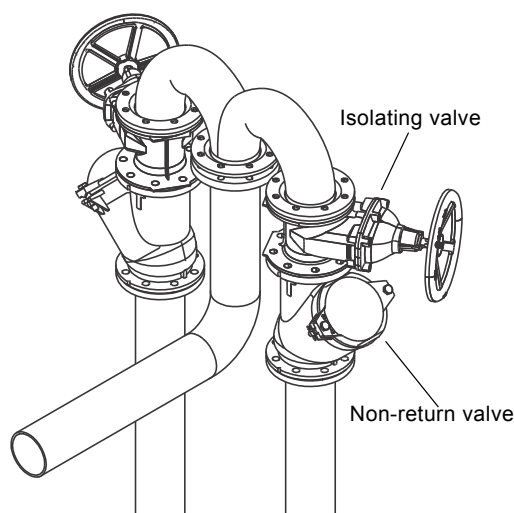


Fig. 31 Gooseneck with valves

Pumping station with valve chamber

1. Open the valve chamber drainage valve by means of the valve handle in the pit in order to empty the sump of the valve chamber. See fig. 32.
2. Close the isolating valves placed inside the valve chamber. See fig. 7.
3. Remove the two screws on the non-return valve cover. See fig. 33.
4. Replace worn-out valve balls and clean the valves inside.
5. Fit the valve cover.
6. Open the isolating valves placed inside the valve chamber. See fig. 7.
7. Close the drainage valve inside the pit. See fig. 32.

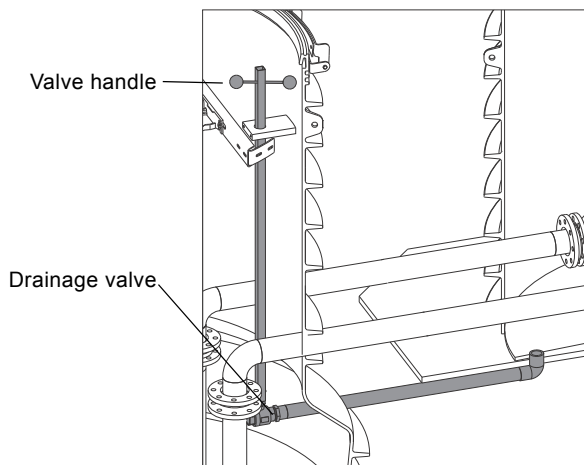


Fig. 32 Draining the valve chamber

Non-return valve

Non-return valve, exploded view

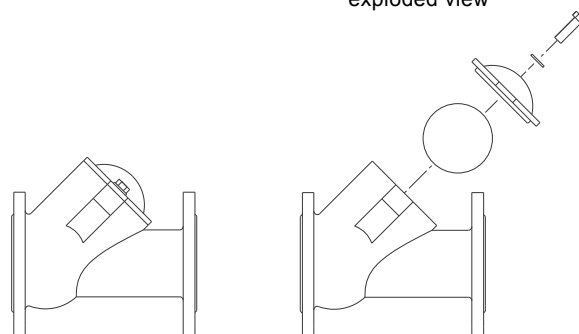


Fig. 33 Non-return valve

8.2 Contaminated pumps



Warning

If a pump has been used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

If Grundfos is requested to service the pump, Grundfos must be contacted with details about the pumped liquid, etc. before the pump is returned for service. Otherwise Grundfos can refuse to accept the pump for service.

Possible costs of returning the pump are to be paid by the customer.

However, any application for service, no matter to whom it may be made, must include details about the pumped liquid if the pump has been used for liquids which are injurious to health or toxic.

Before a pump is returned, it must be cleaned in the best possible way.

Service instruction and service video can be found on www.grundfos.com.

9. Service contract

It is possible to make a service contract with Grundfos.

10. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.
3. If it is not possible to dispose of the product as a complete unit, the pump and installation equipment can be removed from the pit, and the pit can be filled and covered.

Declaration of performance

Declaration of performance

GB:**EC declaration of performance in accordance with Annex III of Regulation (EU) No 305/2011 (Construction Product Regulation)**

1. Unique identification code of the product type:
– EN 12050-1 or EN 12050-2.
2. Type, batch or serial number or any other element allowing identification of the construction product as required pursuant to Article 11(4):
– Pumping stations marked with EN 12050-1 or EN 12050-2 on the nameplate.
3. Intended use or uses of the construction product, in accordance with the applicable harmonised technical specification, as foreseen by the manufacturer:
– Pumping stations for pumping of wastewater containing faecal matter marked with EN 12050-1 on the nameplate.
– Pumping stations for pumping of faecal-free wastewater marked with EN 12050-2 on the nameplate.
4. Name, registered trade name or registered trade mark and contact address of the manufacturer as required pursuant to Article 11(5):
– Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Denmark.
5. NOT RELEVANT.
6. System or systems of assessment and verification of constancy of performance of the construction product as set out in Annex V:
– System 3.
7. In case of the declaration of performance concerning a construction product covered by a harmonised standard:
– TÜV Rheinland LGA Products GmbH, identification number: 0197.
Performed test according to EN 12050-1 or EN 12050-2 under system 3.
(description of the third party tasks as set out in Annex V)
– LGA test report No. 21226935-001.
8. NOT RELEVANT.
9. Declared performance:
The products covered by this declaration of performance are in compliance with the essential characteristics and the performance requirements as described in the following:
– Standards used: EN 12050-1:2001 or EN 12050-2:2000.
10. The performance of the product identified in points 1 and 2 is in conformity with the declared performance in point 9.

DK:**EU-ydeevnedeklaration i henhold til bilag III af forordning (EU) nr. 305/2011 (Byggevareforordningen)**

1. Varetypens unikke identifikationskode:
– EN 12050-1 eller EN 12050-2.
2. Type-, parti- eller serienummer eller en anden form for angivelse ved hjælp af hvilken byggevaren kan identificeres som krævet i henhold til artikel 11, stk. 4:
– Pumpestationer der er mærket med EN 12050-1 eller EN 12050-2 på typeskiltet.
3. Byggevarens tilsigtede anvendelse eller anvendelser i overensstemmelse med den gældende harmoniserede tekniske specifikation som påtænkt af fabrikanten:
– Pumpestationer til pumpning af spildevand med fækalier der er mærket med EN 12050-1 på typeskiltet.
– Pumpestationer til pumpning af fækaliefrit spildevand der er mærket med EN 12050-2 på typeskiltet.
4. Fabrikantens navn, registrerede firmabetejgnelse eller registrerede varemærke og kontaktadresse som krævet i henhold til artikel 11, stk. 5:
– Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danmark.
5. IKKE RELEVANT.
6. Systemet eller systemerne til vurdering og kontrol af at byggevarens ydeevne er konstant, jf. bilag V:
– System 3.
7. Hvis ydeevnedeklarationen vedrører en byggevare der er omfattet af en harmoniseret standard:
– TÜV Rheinland LGA Products GmbH, identifikationsnummer: 0197.
Udført test i henhold til EN 12050-1 eller EN 12050-2 efter system 3 (beskrivelse af tredjepartsopgaverne, jf. bilag V).
– LGA-testrapport nr. 21226935-001.
8. IKKE RELEVANT.
9. Deklareret ydeevne:
De produkter der er omfattet af denne ydeevnedeklaration, er i overensstemmelse med de væsentlige egenskaber og ydelseskrav der er beskrevet i følgende:
– Anvendte standarder: EN 12050-1:2001 eller EN 12050-2:2000.
10. Ydeevnen for den byggevare der er anført i punkt 1 og 2, er i overensstemmelse med den deklarerede ydeevne i punkt 9.

SE:

**EG prestandadeklaration enligt bilaga III till förordning (EU)
nr 305/2011
(byggproduktförordningen)**

1. Produkttypens unika identifikationskod:
– EN 12050-1 eller EN 12050-2.
2. Typ-, parti- eller serienummer eller någon annan beteckning som möjliggör identifiering av byggprodukter i enlighet med artikel 11.4:
– Pumpstationer märkta med EN 12050-1 eller EN 12050-2 på typskylten.
3. Byggproduktens avsedda användning eller användningar i enlighet med den tillämpliga, harmoniserade tekniska specifikationen, såsom förutsett av tillverkaren:
– Pumpstationer för pumpning av avloppsvatten innehållande fekalier märkta med EN 12050-1 på typskylten.
– Pumpstationer för pumpning av fekaliefritt avloppsvatten märkta med EN 12050-2 på typskylten.
4. Tillverkarens namn, registrerade företagsnamn eller registrerade varumärke samt kontaktadress enligt vad som krävs i artikel 11.5:
– Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
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5. EJ TILLÄMPLIGT.
6. Systemet eller systemen för bedömning och fortlöpande kontroll av byggprodukts prestanda enligt bilaga V:
– System 3.
7. För det fall att prestandadeklarationen avser en byggprodukt som omfattas av en harmoniserad standard:
– TÜV Rheinland LGA Products GmbH, identifikationsnummer: 0197.
Utförde provning enligt EN 12050-1 eller EN 12050-2 under system 3.
(beskrivning av tredje parts uppgifter såsom de anges i bilaga V)
– LGA-testprotokoll nr 21226935-001.
8. EJ TILLÄMPLIGT.
9. Angiven prestanda:
Produkterna som omfattas av denna prestandadeklaration överensstämmer med de väsentliga egenskaperna och prestandakraven i följande:
– Tillämpade standarder: EN 12050-1:2001 eller EN 12050-2:2000.
10. Prestandan för den produkt som anges i punkterna 1 och 2 överensstämmer med den prestanda som anges i punkt 9.

EU declaration of performance reference number: 98646710.

Bjerringbro, 26th January 2015



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